

---

# Reading the Router: Plain-Language Profiles of Mixture-of-Experts Specialization

---

Steven Hou  
Independent Researcher

## Abstract

A companion paper argued that AI oversight benefits from rendering a model’s opaque internals into plain English—there, by paraphrasing a chain of thought. Here we apply the same legibility agenda to a different internal: the routing of a Mixture-of-Experts (MoE) model. We extract genuine per-token expert assignments from OLMoE-1B-7B (16 layers, 64 experts, top-8) on a category-labeled corpus, quantify how specialized each expert is, and hand each specialized expert’s distinctive tokens to a small trusted model that writes a one-line, plain-English description of what it does. We find that (i) most experts are balanced (mean specialization 0.19), but (ii) a legible specialized tail exists (2.1% of layer-expert units score  $> 0.5$ ) and (iii) every category has a strongly dedicated expert (peak routing propensity 0.88–0.99 versus a 0.125 uniform baseline). Specialization concentrates in late layers. The plain-language profiles are frequently accurate and human-readable—a “calculus and trigonometric notation” expert, a “Python syntax” expert, a “speech verbs and quotation marks” expert, and shared Romance-language experts. We argue router-level plain-language profiling is a cheap, model-agnostic legibility tool, and we are honest about its limits: specialization is modest, category-defined, correlational, and the descriptions are only as good as the labeller.<sup>1</sup>

## 1 Introduction

Mixture-of-Experts (MoE) models replace a dense feed-forward block with many “experts” and a router that sends each token to a sparse top- $k$  subset (Shazeer et al., 2017; Fedus et al., 2022). This buys parameter count without proportional compute, and powers strong open models (Jiang et al., 2024; Muennighoff et al., 2024). But the router is opaque: a long-standing question is what each expert specializes in, and answers have been mixed—Jiang et al. (2024) report little obvious domain structure in Mixtral, while Muennighoff et al. (2024) find clearer vocabulary and domain specialization in OLMoE.

Our companion paper studied a different opaque internal—a model’s chain of thought—and made it legible for oversight by forcing it through a plain-English paraphrase. We pursue the same agenda for routing: render the expert structure into plain English so a human can read it. Concretely, we (1) extract real per-token routing from OLMoE-1B-7B, (2) measure each expert’s specialization over a labeled corpus, and (3) profile specialized experts in one plain sentence each, using a small trusted model as the labeller. The result is a human-readable map of part of the router.

---

<sup>1</sup>Code and data: local artifact at `~/moe_legibility/`. Single 7B-MoE, 1,541-token corpus proof of concept; Section 5 states scope.

Contributions. (i) A simple, exact routing-extraction hook and a specialization metric for MoE experts (Section 2). (ii) An empirical map of OLMoE specialization: modest on average, concentrated in late layers, with a strongly-dedicated expert per category (Section 3). (iii) Plain-language expert profiles generated from genuine routing, with an honest account of when they are accurate and when they are not (Section 3.1, 5).

## 2 Method

Model and routing extraction. We use OLMoE-1B-7B-Instruct (4-bit) (Muennighoff et al., 2024): 16 MoE layers, 64 experts each, top-8 routing, 1.3B active parameters. We wrap each MoE block’s gate to record, for every token, the indices of its top-8 experts at every layer—the exact selections the model makes, not an approximation.

Corpus and specialization. We route a corpus of 1,541 tokens spanning eight categories (Python, math, English, French, Spanish, JSON, dialogue, Chinese). Let  $c_{l,e,k}$  count tokens of category  $k$  that select expert  $e$  at layer  $l$ , and  $n_k$  the tokens in category  $k$ . The routing propensity is  $p_{l,e,k} = c_{l,e,k}/n_k$  (expected selections per token; a uniform router gives  $8/64 = 0.125$ ). Normalising  $p$  over categories to  $q$ , the specialization of a layer-expert unit is  $s_{l,e} = 1 - H(q_{l,e})/\log 8 \in [0, 1]$  ( $0$  = balanced across categories,  $1$  = one category).

Plain-language profiling. For each specialized expert we collect its distinctive tokens—those whose frequency among its selections most exceeds their corpus frequency (lift)—and give them, with the dominant category, to a small trusted model (llama3.2:3b) prompted to name the token pattern in  $\leq 8$  words. This is the legibility step: it converts a list of token indices into a sentence a person can read.

## 3 Results

Specialization is modest but structured. Mean specialization is 0.19: most experts fire across many categories (Fig. 3), echoing the “little obvious specialization” reported for Mixtral. But the distribution has a legible tail—2.1% of the 1,024 layer-expert units exceed 0.5—and this tail is where plain-language reading pays off. Specialization is not uniform with depth: it rises in late layers (Fig. 2), with the most specialized experts concentrated in layers 13–15.

Every category has a dedicated expert. Although the average expert is balanced, the converse view is striking: for every category there exists an expert it selects with propensity 0.88–0.99—seven to eight times the uniform baseline (Fig. 4). The category $\times$ expert map (Fig. 1) shows the structure directly: bright vertical bands are specialized experts, French and Spanish light up the same experts (shared Romance-language routing), structured text (Python, math, JSON) shares others, and English is spread broadly—it is the model’s “default” register.

### 3.1 Plain-language expert profiles

Table 1 lists profiles generated from genuine routing. Several are crisp and verifiable against their evidence tokens: a calculus expert (`ln`, `integral`, `cos`, `=`, `/`), a Python-syntax expert (`def`, `return`, `(`, `=`), a speech-verb/quotation expert (`said`, `asked`, `replied`), and a French-function-word expert. These are exactly the kind of human-readable handles the legibility agenda seeks: one can look at the router and say, in plain English, what a part of it is for.

## 4 Discussion

Read together with the companion paper, a theme emerges: opaque model internals—a chain of thought, a routing table—can be partially rendered into plain English for human oversight, cheaply and without retraining. For MoE, the win is real but bounded: the router

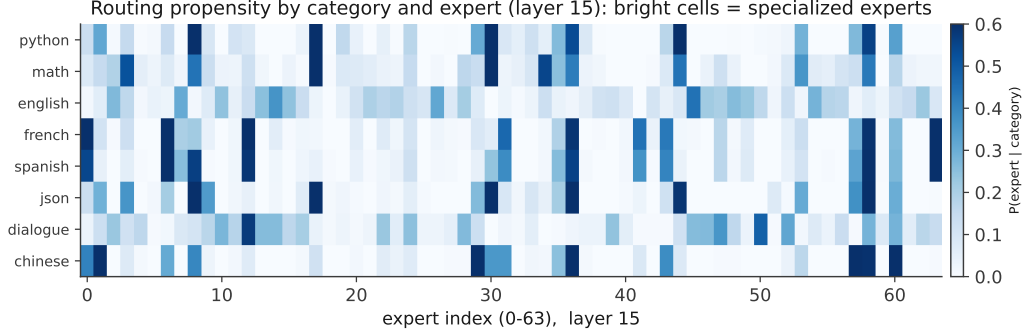


Figure 1: Genuine routing propensity  $P(\text{expert} \mid \text{category})$  at the most specialized layer. Bright cells are specialized experts; French/Spanish share experts, structured text (Python/math/JSON) shares others, English is diffuse.

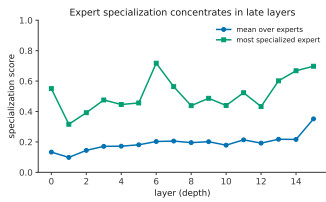


Figure 2: Specialization vs. depth.

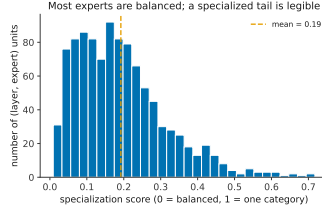


Figure 3: Specialization distribution.

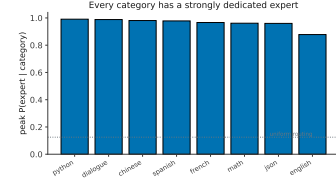


Figure 4: Peak expert per category.

is not a tidy set of domain modules (Jiang et al., 2024’s caution holds—mean specialization is low), yet it does contain a legible minority of experts that a person can name, and its coarse structure (late-layer specialization, shared Romance-language experts, a diffuse English default) is itself interpretable. Plain-language profiling turns that minority into oversight-ready handles: “this expert handles arithmetic notation,” “this one handles quoted speech.”

## 5 Limitations

We state scope plainly. (1) Small probe corpus (1,541 tokens, eight hand-chosen categories): specialization scores are category-defined—a different taxonomy would relabel experts, and rare patterns are missed. (2) Single model, OLMoE-1B-7B at 4-bit; larger MoEs (Mixtral, DeepSeek) may differ. (3) Correlational, not causal: we observe routing, not expert ablations, so a profile says “fires on,” not “is necessary for.” (4) Labeller-dependent: descriptions come from a 3B model and can be wrong or over-confident (we saw it first mislabel dialogue experts as “conversational AI”); the evidence tokens, not the sentence, are the ground truth. (5) Common-token noise: pronoun/punctuation experts are harder to name distinctively than domain experts.

## 6 Conclusion

We read part of an MoE router in plain English. Extracting genuine routing, measuring specialization, and profiling specialized experts with a trusted small model yields human-readable descriptions of what experts do—a calculus expert, a Python expert, a quoted-speech expert—alongside an honest picture of how modest and structured MoE specialization really is. As with paraphrasing a chain of thought, the contribution is not that internals become fully legible, but that a useful, oversight-relevant slice of them can be, cheaply.

Table 1: Plain-language profiles of specialized OLMoE experts, generated by a trusted small model from genuine routing. “L/E” = layer/expert;  $s$  = specialization.

| L/E   | $s$  | plain-language profile                | evidence tokens                              |
|-------|------|---------------------------------------|----------------------------------------------|
| 13/20 | 0.60 | calculus and trigonometric notation   | <code>ln, integral, equals, cos, *, /</code> |
| 0/52  | 0.55 | Python syntax (def, return, brackets) | <code>return, lines, =, (</code>             |
| 7/15  | 0.56 | algebraic expressions and symbols     | <code>equals, of, is</code>                  |
| 15/39 | 0.69 | speech verbs and quotation marks      | <code>replied, asked, said</code>            |
| 15/26 | 0.61 | French articles and conjunctions      | <code>la, light, the, and</code>             |
| 15/50 | 0.70 | personal pronouns and imperatives     | <code>I, you, You, me</code>                 |
| 6/51  | 0.72 | CJK punctuation and dialogue marks    | <code>," , ."</code>                         |
| 15/21 | 0.56 | negations and interrogatives          | <code>the, no, a</code>                      |

## References

- W. Fedus, B. Zoph, and N. Shazeer. Switch transformers: scaling to trillion parameter models with simple and efficient sparsity. JMLR; arXiv:2101.03961, 2022.
- A. Jiang et al. Mixtral of experts. arXiv:2401.04088, 2024.
- N. Muennighoff et al. OLMoE: open mixture-of-experts language models. arXiv:2409.02060, 2024.
- N. Shazeer et al. Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. ICLR; arXiv:1701.06538, 2017.